

Álvaro Martínez Higuera

Computer Science & Engineering Student · Software Development

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3rd-year Computer Science and Engineering student at the University of Alicante, focusing on artificial intelligence. Hands-on experience building and running full production applications of his own, both web-based (backend, frontend, infrastructure and security) and cross-platform desktop/mobile. Looking for a summer 2027 internship, available 15 June to 6 September, to contribute and keep learning.

PROJECTS

PredictMotion 2025 – present

Personal project · predictmotion.com

- Football probability website (12 leagues and UEFA competitions) powered by a custom Python Monte Carlo simulation engine, validated with a backtest of over 9,500 real matches.
- Fully self-managed infrastructure end to end: Linux server on Oracle Cloud, continuous deployment, Cloudflare, monitoring and alerting.
- User authentication system (Sign in with Google), SQLite database, encrypted backups and GDPR compliance.
- Conducted own security audit and fixed vulnerabilities; complete UI redesign for web and mobile.

Insurance Portfolio Manager 2025 – present

Personal project · desktop (Windows) and Android app

- Application for managing an insurance policy portfolio (clients, expirations, renewal alerts, calendar), built with Svelte 5 using a hexagonal architecture (domain / application / infrastructure) to share business logic between desktop (Electron) and mobile (Capacitor).
- Cloud backend with Supabase (Postgres, passwordless authentication, Row Level Security) with local caching and offline mode.
- Automated release pipeline: signed Windows installer and Android APK builds via GitHub Actions, with auto-update on both platforms.

EDUCATION

2023 – 2028 (EXPECTED)

B.Sc. in Computer Science and Engineering

Universidad de Alicante (UA)

Current coursework: Algorithm Analysis and Design, Computer Architecture, Software Systems Design, Programming Languages and Paradigms, Computer Networks, Intelligent Systems.

FALL SEMESTER 2026/2027

Erasmus Exchange – Technical University of Sofia

Sofia, Bulgaria

TECHNICAL SKILLS

LANGUAGES

Python, TypeScript/JavaScript, PHP, Swift, HTML/CSS, MIPS Assembly

AREAS

Algorithms, computer architecture, software design (SOLID, GRASP, GoF patterns, UML), networking, artificial intelligence (neural networks, Bayesian networks, CSPs, fuzzy logic, boosting)

TOOLS

Git/GitHub, Linux, Svelte, Electron, Capacitor, Supabase, Caddy, Cloudflare, Oracle Cloud

LANGUAGES

Spanish: native

English: B2

ADDITIONAL INFORMATION

Personal portfolio: alvaromartinezh.github.io

Available full-time from 15 June to 6 September 2027 (approx. 12 weeks).